25901647

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION

APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCCFS100 - WHITE HATHACKING AND TOOLS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks : 35**

Part A

Short Answer Type Questions

Answer any 10 questions

Each question carries 1 marks

1. Identify the Indian act that governs cyber crimes and electronic commerce. [U] / [CO1]

The Information Technology Act, 2000 – Identification-1 mark

2. Identify the cyber crime that involves creating fake websites to deceive users. [U] / [CO1]

Phishing – Identification-1 mark

3. Describe Reconnaissance. [U] / [CO1]

Reconnaissance is the information-gathering stage of ethical hacking, where you collect data about the target system before launching an attack.

Explanation – 1 mark

4. Explain the punishment for hacking under the IT Act. [U] / [CO1]

Section 66 - Imprisonment up to 3 years Or fine up to ₹5 lakh Or both.

Explanation – 1 mark

5. Name the three types of hacking and state one key feature for each. [U] / [CO1]

* White Hat Hacking (Ethical Hacking) - Key Feature: Performed legally to identify and fix security vulnerabilities.

* Black Hat Hacking - Key Feature: Performed illegally with malicious intent such as stealing data or causing damage.

* Grey Hat Hacking - Key Feature: Accesses systems without permission but usually not for personal gain; may disclose vulnerabilities later.

Listing 3 types – ½ mark. Stating key feature – ½ mark

6. Describe the purpose of hacking from a security perspective. [U] / [CO1]

Main purposes include finding vulnerabilities, Testing System Security, Preventing Data Breaches, Improving Security Measures, Ensuring Compliance, Creating Security Awareness etc.

Explanation – 1 mark

7. Explain the term 'digital signature' refer to in cyber law. [U] / [CO1]

A digital signature is an electronic method used to verify the authenticity, integrity, and identity - 1 mark

8. Why do attackers clear logs after post-exploitation? [U] / [CO1]

Attackers clear logs after post-exploitation to hide their activities and avoid detection or forensic investigation. – 1 mark

9. Describe exploitation. [U] / [CO1]

Exploitation is the process of taking advantage of a vulnerability in a computer system, network, or application to gain unauthorized access or perform malicious actions. – 1 mark

10. Define Packet inspection firewall. [U] / [CO2]

A packet inspection firewall is a firewall that examines data packets passing through a network and allows or blocks them based on security rules. – 1 mark

11. Define Vulnerability. [U] / [CO2]

A vulnerability is a weakness or flaw in a computer system, network, or software that can be exploited by an attacker to cause harm or gain unauthorized access. – 1 mark

12. Describe malware attack. [U] / [CO2]

A malware attack is a cyber attack in which malicious software is used to damage, disrupt, steal data from, or gain unauthorized access to a computer system or network. – 1 mark

[1x10 = 10]

Part B

Short Essay Type Questions

Answer any 3 questions

Each question carries 5 marks

13. What are the main components of reporting? [U] / [CO1]

Executive Summary, Technical Report, Findings and Vulnerabilities, Risk Assessment, Recommendations/Remediation, Supporting Evidence and Conclusion.

Listing out components – 2 marks

Detailed explanation – 3 marks

14. Discuss how Ethical Hackers can get Hands-On Practice Gaining Access. [U] / [CO1]

Ethical hackers can get hands-on practice gaining access by using legal and controlled environments to learn hacking techniques and security testing methods. - Virtual Labs, Practice Platforms, Capture The Flag (CTF) Challenges, Penetration Testing Tools,

Detailed explanation – 5 marks

15. Discuss how to prevent cyber attack. [U] / [CO2]

Cyber attacks can be prevented by adopting proper security measures and safe online practices - Use Strong Passwords, Install Antivirus and Firewall, Regular Software Updates, Use Secure Networks, Access Control and Authentication, Security Awareness Training, Regular Security Audits etc.

Detailed explanation – 5 marks

16. Differentiate Phishing and Spoofing. [U] / [CO2]

Phishing is a cyber attack used to steal sensitive information from users. Spoofing is a technique used to disguise/fake the sender's identity. Spoofing hides the attacker's identity, and phishing tries to trick users into giving important information.

Phishing and spoofing definition – 2 marks

Explain differences – 3 marks

[5x3 = 15]

Part C

Essay Type Questions

Answer any 1 question

Each question carries 10 marks

17. Explain the stages of ethical hacking. [U] / [CO1]

Five stages - Reconnaissance, Scanning, Gaining access, Maintaining access, Clearing tracks.

Explain each stage – 2 marks each. (2x5=10 marks)

18. Explain the difference between man -in -the middle attack, credential stuffing, bruteforce attack, dictionary attack. [U] / [CO2]

Cyber attacks introduction – 2 marks

Explain each attack – 2 marks each (2x4=8 marks)

[10x1 = 10]